

Chapter 2

Coin flips $\rightarrow$ 5 coins

H H T T H $\leftarrow$ microstate

T H H H H $\leftarrow$ microstate

different microstate $\rightarrow$ H H H T T

3 heads $\leftarrow$ macrostate

how many microstates are in a macrostate?

$\hookrightarrow$ multiplicity

$$\Omega(N, n) = \frac{N!}{n!(N-n)!}$$

$\uparrow$
of heads

$N \leftarrow$ total number of coins
 $n \leftarrow$ number of heads

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

Ex: 5 coins

$$\Omega(5) = \frac{5!}{5!(5-5)!} = \frac{5!}{5! \underbrace{0!}} = 1$$

$0! = 1$

$$\Omega(4) = \frac{5!}{\underbrace{4!(5-4)!}} = \frac{5 \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1}{\cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1} = 5$$

$1! = 1$

$$x! = x \cdot (x-1)!$$

$$\Omega(3) = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{120}{6 \cdot 2} = 10$$

10 atoms - each atom can have either 0 or 1 energy units.

So, how many combinations are there of arranging 4 units of energy?

$$\Omega(N, n) = \frac{N!}{n!(N-n)!}$$

What if more than one energy can be on each atom?

$$\Omega(N, q) = \frac{(q + N - 1)!}{q!(N-1)!}$$

↑
N atoms

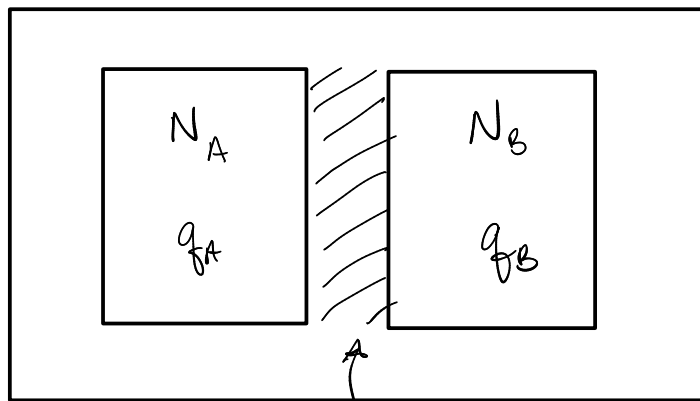
↑
number of energy packets to share

○ ○ ○ ● ○ ○ ○ ○ ○ ○ ○ ○ ○ ○
1e 2e 1e

← one particular microstate in the macrostate of 4 energies among 10 atoms

This model of how materials behave is known as an Einstein Solid.

Two Systems - these can exchange energy



isolated from surroundings

energy can flow from one side to the other but not particles

$$\Omega(N, q) = \frac{(q + N - 1)!}{q! (N - 1)!}$$

$$= \frac{(5 + 3 - 1)!}{5! (3 - 1)!} = 21$$

$\textcircled{1} \quad \textcircled{2} \quad \textcircled{0}$
 $1 \quad 2 \quad 0$
 macrostate of 3 quanta

$\textcircled{1} \quad \textcircled{1} \quad \textcircled{1}$
 $1 \quad 1 \quad 1$
 macrostate of 3 quanta

$\textcircled{2} \quad \textcircled{0} \quad \textcircled{1}$

$\textcircled{3} \quad \textcircled{0} \quad \textcircled{0}$
 3

$\textcircled{1} \quad \textcircled{0} \quad \textcircled{0}$
 macrostate of 1 quanta

$\textcircled{3} \quad \textcircled{0} \quad \textcircled{2}$
 macrostate of 5

$$\Omega_A(3, 1) = 3$$

$$\Omega_B(3, 5) = 21$$

what is the total multiplicity of these macrostates?

$$\Omega_{\text{total}} = \Omega_A \cdot \Omega_B$$

$$= 63$$

↑ What is this the same as?

$$\mathcal{L}(6,6) = \frac{(6+6-1)!}{6!(6-1)!}$$

$$= \frac{11!}{6!5!} = 462$$

↑
I had some extra information before!

Fundamental Assumption of Statistical Mechanics

↳ all microstates are possible and equally probable

But. That does not mean that every microstate will occur.

Not all microstates are equally probable.

Large Number $\rightarrow$ addition of small #'s
does not matter

$$10^{23} + 100 = 10^{23}$$

Very Large Numbers $\rightarrow$ multiplication
of small or large #'s does not matter

$$10^{10^{23}} \times 10^{23} = 10^{10^{23} + 23} \approx 10^{10^{23}}$$

$$\Omega(N, q) = \frac{(q+N-1)!}{q! (N-1)!} \approx \frac{(q+N)!}{q! N!}$$

$$\ln \Omega = \ln[(q+N)!] - \ln(q!) - \ln(N!) \\ \begin{aligned} &\rightarrow (q+N) \ln(q+N) - (q+N) \\ &\quad \rightarrow q \ln q - q \\ &\quad \rightarrow N \ln N - N \end{aligned}$$

Stirling's Approximation

$$N! \approx N^N e^{-N} \sqrt{2\pi N} \approx N^N e^{-N} N^{1/2} \\ \approx N^{N+1/2} e^{-N}$$

$$N! \approx N^N e^{-N} = \frac{N^N}{e^N}$$

$$\ln N! \approx \ln(N^N) - \ln(e^N)$$

$$\ln N! \approx N \ln N - N$$

$$\ln \Omega = (q+N) \ln(q+N) - \cancel{q} - \cancel{N} - q \ln q + \cancel{q} - N \ln N + \cancel{N}$$

$$\ln \Omega = (q+N) \ln(q+N) - q \ln q - N \ln N$$

what is Ω in the high temperature limit?

$$q \gg N$$

$$\ln \Omega = (\overbrace{q+N}^{q+N}) \ln(q+N) - q \ln q - N \ln N$$

$$= q \ln(q+N) + N \ln(q+N) - q \ln q - N \ln N$$

$$\underbrace{\quad}_{\downarrow} = \ln\left(q\left(1 + \frac{N}{q}\right)\right)$$

$$= \ln q + \ln\left(1 + \frac{N}{q}\right)$$

$$\underbrace{\quad}_{\rightarrow \text{a bit like}} \ln(1+x) \approx x$$

$$= \frac{N}{q}$$

$$\rightarrow = \ln q + \frac{N}{q} \approx \ln(q+N)$$

$$\ln \Omega = \cancel{q \ln q} + \cancel{q} \cdot \frac{N}{\cancel{q}} + N \ln q + \frac{N^2}{q} - \cancel{q \ln q} - N \ln N$$

$$\begin{aligned} \ln \Omega &= N \ln q - N \ln N + N + \frac{N^2}{q} \\ &= N \ln \left(\frac{q}{N} \right) + N + \underbrace{\frac{N^2}{q}}_{\text{small}} \end{aligned}$$

$$\ln \Omega(q \gg N) = N \ln \left(\frac{q}{N} \right) + N$$

$$\begin{aligned} \Omega(q \gg N) &= e^{N \cdot \ln \left(\frac{q}{N} \right) + N} = e^{N \cdot \ln \left(\frac{q}{N} \right)} \cdot e^N \\ &= \left[e^{\ln \left(\frac{q}{N} \right)} \right]^N \cdot e^N \\ &= \left(\frac{q}{N} \right)^N \cdot e^N \end{aligned}$$

work problem 2.18.
2.22.

$$\Omega(q \gg N) = \left(\frac{e \cdot q}{N} \right)^N$$

← multiplicity
of an Einstein
solid at
high temperature

Entropy

Any large system in equilibrium will be found in the macrostate w/ the greatest multiplicity.

2nd law of Thermodynamics
Multiplicity tends to increase.

$$S = k_B \ln \Omega$$

Entropy $\rightarrow$ the log of the number of ways of arranging our system.

Ex: an Einstein solid at high temp. $q \gg N$

$$N = 10^{23} \quad q = 10^{25}$$

$$\Omega(q \gg N) = \left(\frac{e \cdot q}{N} \right)^N$$

$$S = k_B \ln \left[\left(\frac{e \cdot g}{N} \right)^N \right]$$

$$S = k_B \cdot N \cdot \ln \left(\frac{e \cdot g}{N} \right)$$

$$= k_B \cdot N \cdot \left(\ln \left(\frac{g}{N} \right) + \underbrace{\ln(e)}_1 \right)$$

$$S = k_B \cdot N \cdot \left(\ln \left(\frac{g}{N} \right) + 1 \right)$$

$$N = 10^{23} \quad g = 10^{25}$$

$$S = 1.38 \cdot \cancel{10^{-23}} \frac{\text{J}}{\text{K}} \cdot \cancel{10^{23}} \cdot \left(\ln \left(\frac{10^{25}}{10^{23}} \right) + 1 \right)$$

$$S = 1.38 \frac{\text{J}}{\text{K}} \cdot (4.6 + 1)$$

$$S = 7.7 \frac{\text{J}}{\text{K}}$$

For a composite system, $\Omega_{\text{total}} = \Omega_A \Omega_B$

$$S_{\text{total}} = k_B \ln \Omega_{\text{total}} = k_B \ln \Omega_A \Omega_B$$

$$= k_B \ln \Omega_A + k_B \ln \Omega_B$$

$$S_{\text{total}} = S_A + S_B$$

Any large system in equilibrium
will be found in the macrostate
w/ the greatest ~~multiplicity~~.
entropy

2nd law of Thermodynamics

~~Multiplicity~~ tends to increase.

Entropy

spontaneous process

2.29 + 2.30

Total HW: 2, 18, 22, 29, 30 Due: Wednesday October 7th

- grade tests
- post solutions to optional HW

Chapter 3 → define temperature → thermal equilibrium

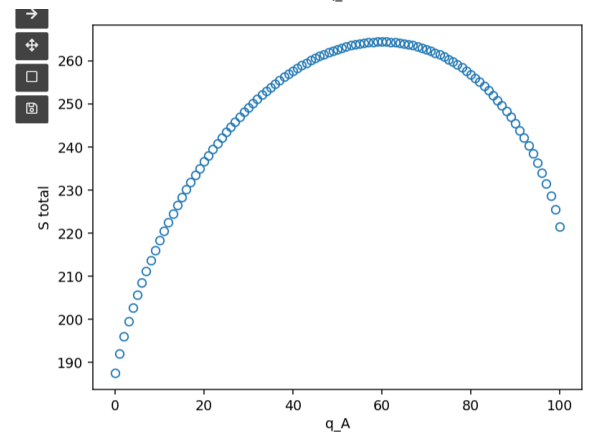
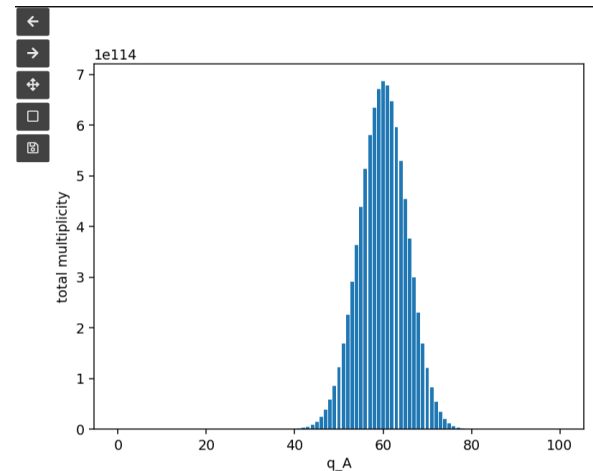
For an Einstein solid

$$\frac{\partial S_{\text{total}}}{\partial q_A} = 0$$

$$\frac{\partial S_{\text{total}}}{\partial U_A} = 0$$

$$\frac{\partial (S_A + S_B)}{\partial U_A} = 0$$

$$S_{\text{total}} = S_A + S_B$$



$$\frac{\partial S_A}{\partial U_A} + \frac{\partial S_B}{\partial U_A} = 0$$

$$\frac{\partial S_A}{\partial U_A} - \frac{\partial S_B}{\partial U_B} = 0$$

$$\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$$

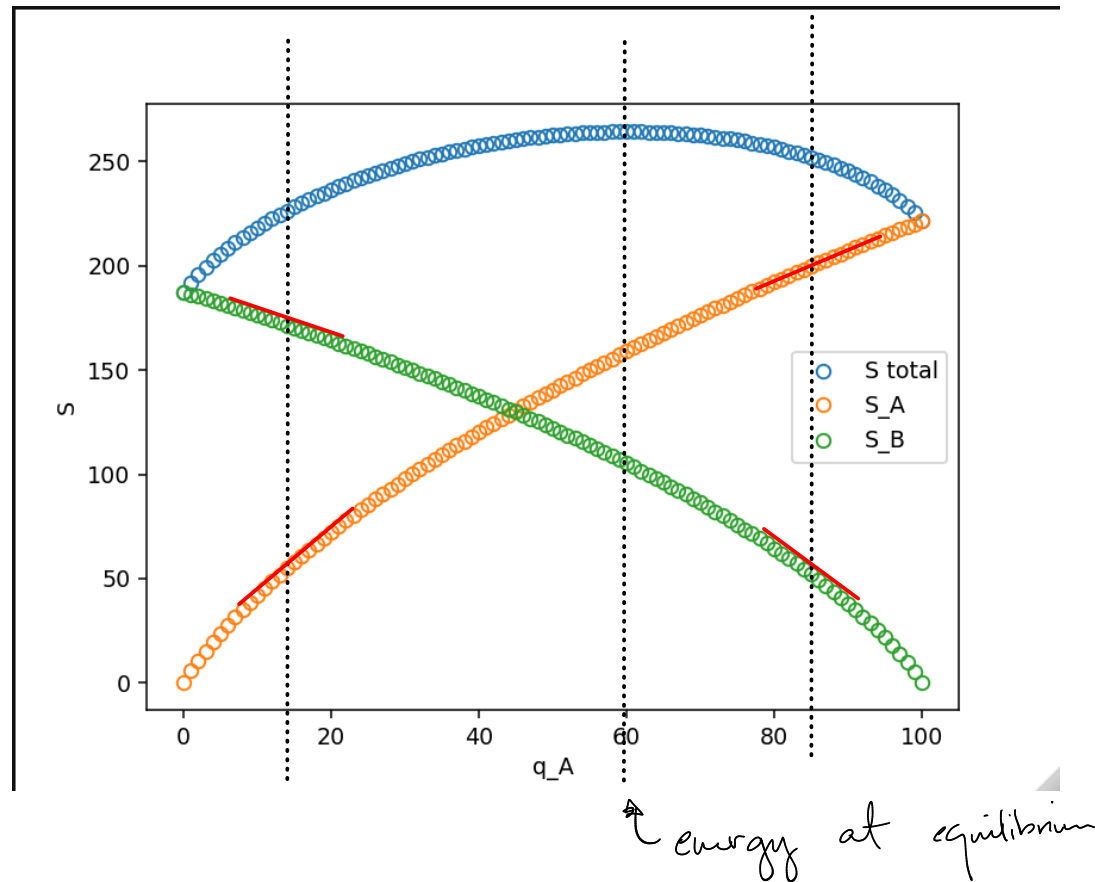
$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,V}$$

$$T = \left(\frac{\partial S}{\partial U} \right)^{-1}_{N,V}$$

$$U_A + U_B = U$$

$$U_B = U - U_A$$

$$dU_B = -dU_A$$



Let's apply what we have learned
to an Einstein solid at high temp:

$$\Omega(q \gg N) = \left(\frac{eq}{N}\right)^N$$

$$S = k_B \ln \Omega$$

$$S = Nk_B \ln\left(\frac{eq}{N}\right)$$

$$S = Nk_B \left(1 + \ln\left(\frac{q}{N}\right)\right)$$

$$S = Nk_B \left(1 + \ln\left(\frac{U}{\epsilon N}\right)\right)$$

$$= Nk_B + Nk_B \ln(U) - Nk_B \ln(\epsilon N)$$

$$U = q \cdot \epsilon$$

of
energy
packets

energy of
each packet

Very naive prediction

$$U = \frac{f}{2} \cdot N \cdot k_B \cdot T$$

$$f = 2$$

Einstein solid

1D kinetic

1D spring
potential
energy

$$U = Nk_B T$$

$$T = \frac{U}{Nk_B}$$

$$\frac{1}{T} = \left(\frac{dS}{dU}\right)_{nv} = \frac{Nk_B}{U} \Rightarrow T = \frac{U}{Nk_B} \Rightarrow$$

$$U = Nk_B T$$

$U = \underline{\text{Power}} \cdot \text{time}$

measure the temp

$$T = \left(\frac{\partial S}{\partial U} \right)^{-1}_{N,V}$$

$$P = I \cdot V \\ = \frac{V^2}{R} = I^2 \cdot R$$

So let's review our process so far:

1. Find Ω in terms of N, U, V, P } could be impossible!
 - We used combinatorics, approximations, QM

2. $S = k_B \ln \Omega$ to get entropy

3. $T = \left(\frac{\partial S}{\partial U} \right)^{-1}_{N,V}$

4. solve step 3 equation for U
 as a function of T

statistical mechanics
 gives a alternative
 approach to arriving
 at step 4.

5. $C_V = \frac{\partial U}{\partial T}$

2.8

macrostates $\rightarrow$ # of energies 20 total + 1 empty state = 21 macrostates

Ω $\leftarrow$ number of microstate
in 1 particular macrostate

$$\Omega(N, q) = \frac{(q + N - 1)!}{q! (N - 1)!}$$

$$\frac{(20 + 20 - 1)!}{20! (19)!} = \frac{39!}{20! 19!} = 6.9 \cdot 10^{10}$$

total $N = 20$ total $q = 20$

all energy in 10 clams

$$\frac{(20 + 10 - 1)!}{20! (9)!} = 2.3 \cdot 10^{22}$$

probability $\uparrow$
all in A

$$= \frac{\# \text{ in one state}}{\text{all the } \# \text{ in all states}} = \frac{\Omega(10, 20) \cdot \Omega(10, 0)}{\Omega_{\text{All}}(20, 20)} = \frac{1 \cdot 10^7}{6.9 \cdot 10^{10}} = \underline{\underline{1.45 \cdot 10^{-4}}}$$

$$\frac{(10 + 10 - 1)!}{10! 9!} = \frac{19!}{10! 9!}$$

probability $\uparrow$
half in A

$$= \frac{\Omega(10, 10) \Omega(10, 10)}{6.9 \cdot 10^{10}} = \frac{8.5 \cdot 10^9}{6.9 \cdot 10^{10}} = 0.123 \quad \checkmark$$

0.124
 $\uparrow$
probability

